

Decoding olfactory EEG signals for different odor stimuli identification using wavelet-spatial domain feature

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ABSTRACT

Background: Decoding olfactory-induced electroencephalography (olfactory EEG) signals has gained significant attention in recent years, owing to its potential applications in several fields, such as disease diagnosis, multimedia applications, and brain-computer interaction (BCI). Extracting discriminative features from olfactory EEG signals with low spatial resolution and poor signal-to-noise ratio is vital but challenging for improving decoding accuracy.

New methods: By combining discrete wavelet transform (DWT) with one-versus-rest common spatial pattern (OVR-CSP), we develop a novel feature, named wavelet-spatial domain feature (WSDF), to decode the olfactory EEG signals. First, DWT is employed on EEG signals for multilevel wavelet decomposition. Next, the DWT coefficients obtained at a specific level are subjected to OVR-CSP for spatial filtering. Correspondingly, the variance is extracted to generate a discriminative feature set, labeled as WSDF.

Results: To verify the effectiveness of WSDF, a classification of olfactory EEG signals was conducted on two data sets, i.e., a public EEG dataset ‘Odor Pleasantness Perception Dataset (OPPD)’, and a self-collected dataset, by using support vector machine (SVM) trained based on different cross-validation methods. Experimental results showed that on OPPD dataset, the proposed method achieved a best average accuracy of 100% and 94.47% for the eyes-open and eyes-closed conditions, respectively. Moreover, on our own dataset, the proposed method gave a highest average accuracy of 99.50%.

Comparison with existing methods: Compared with a wide range of EEG features and existing works on the same dataset, our WSDF yielded superior classification performance.

Conclusions: The proposed WSDF is a promising candidate for decoding olfactory EEG signals.

1. Introduction

Olfaction, as one of the basic sensory functions for human beings, is far superior to other senses in terms of the number of different physical stimuli that it can distinguish (Morrison, 2014). A recent study reports that healthy humans are capable of discriminating more than 1 trillion olfactory stimuli through an olfactory system with hundreds of different olfactory receptors (Morrison, 2014). The processing of olfactory information in the human brain is an extremely complex process (Tromelin, 2016). Accordingly, many biologists and neuroscientists have tried to study that how the brain perceives odors (Arzi, 2011; Howard et al., 2009).

In recent years, with the rapid development of brain science, computer science, and artificial intelligence, researchers (Abbasi et al., 2020; Kroupi et al., 2016; Lötsch et al., 2019; Saha et al., 2014) have begun to

decode electroencephalogram (EEG) signals induced by different olfactory stimuli, to understand the brain process of odor perception. The research on olfactory-induced EEG (olfactory EEG) signals has numerous potential applications. For example, in clinical medicine (Krbot Skorić et al., 2015), olfactory EEG signals can assist to evaluate the subject's smelling ability. For immersive multimedia environments (Raheel et al., 2019), incorporating olfactory information into a multimedia system is helpful to enhance user multimedia experience. In brain-computer interaction (BCI), the olfactory EEG-based BCI system (Iacoviello et al., 2016) can become an innovative communication tool for disabled people.

Decoding olfactory EEG signals can be considered as a pattern recognition task (Lötsch et al., 2019), which generally includes feature extraction (FE), feature selection and classification. Herein, FE is thought to a key task, since designing suitable EEG features, which can

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effectively characterize odors, contributes significantly in recognizing different olfactory stimuli. However, the collected EEG signals are non-stationary, and usually have low signal-to-noise ratio (SNR) and high levels of crosstalk from different areas of the brain. Such issues pose a primary technical challenge in extracting odor-related features from olfactory EEG signals. Correspondingly, to address such problems, some FE methods have been proposed, including time-domain, frequency-domain, time-frequency-domain, and nonlinear analysis methods.

In time-domain analysis, Aydemir et al. (2018) calculate mean, kurtosis, skewness, derivative, sum of derivatives and singular value decomposition to represent the EEG signals, and propose a subject binary spice odor classification method. Recently, Aydemir (2020) tests the band power (BP), statistical data (kurtosis, skewness, standard deviation, variance, and mean), Hjorth parameters (Hjorth), and autoregressive parameters (AR) features, and points out that the olfactory response of the human brain can be reliably used for odor and subject identifications. For frequency-domain analysis, based on the power spectral density (PSD) feature of EEG signals recorded at different electrodes, Hou et al. (2020b) propose a trapezoid difference-based electrode sequence hashing (TDESH) approach to recognize different olfactory stimuli. In a more recently reported work, based on PSD features of EEG signals, a double-square-based electrode sequence learning method is developed to identify different concentrations of the same odor (Hou and Meng, 2021). In another work, Zhang et al. (2019) employ differential entropy (DE) of EEG signals to distinguish the five different odors. In time-frequency-domain analysis, Aydemir (2017) proposes to use continuous wavelet transform coefficient features (CWTs) for distinguishing different olfactory stimuli. Lastly, for nonlinear analysis methods, four nonlinear features, i.e., Higuchi's fractal dimension (HFD), largest Lyapunov exponent (LLE), approximate entropy (ApEn) and Lempel-Ziv complexity (LZC), are introduced by Ezzatdoost et al. (2020) to classify four olfactory stimuli from recorded EEG signals. Omam et al. (2020, 2021) use the fractal dimension, Shannon entropy, and Hurst exponent of both EEG and Galvanic Skin Response (GSR) signals to decode the coupling between human brain and skin activities, in response to different olfactory stimuli.

Taking a closer look at the above stated examples from literature, it can be noted that most of FE methods utilized in these studies are simply an extension of the single-channel EEG signal FE methods, while disregarding the spatial characteristics of multi-channel EEG signals. To be specific, these FE methods first extract features from the EEG signal of each channel separately, and then directly connect the obtained features in series to form a feature vector. Thus, they cannot capture the spatial distribution pattern to different tasks from multi-channel EEG signals, which likely leads to a degradation in classification performance. Therefore, designing a FE method that can mining abundant and comprehensive odor-related information from multi-channel EEG signals is highly valuable for effectively distinguishing different olfactory stimuli.

Common spatial pattern (CSP), as a spatial filter, is a commonly used FE method for binary classification of motor-imagery (MI) EEG signals (Aggarwal and Chugh, 2019; Mishuhina and Jiang, 2021). The one-versus-rest CSP (OVR-CSP) algorithm (Wu et al., 2005) is an extension of CSP to a multi-class case. Both of them can extract spatial distribution components of each class from multi-channel MI EEG signals. Lately, CSP has been illustrated to be extremely effective in EEG-based emotion recognition (Basar et al., 2020) and seizure onset detection (Li et al., 2021). Nevertheless, both CSP and OVR-CSP algorithms have not been investigated for decoding olfactory EEG signals yet.

Discrete wavelet transform (DWT), as one of time-frequency analysis methods, can simultaneously capture the time location and frequency of signals. In the literature of EEG-based classification, DWT coefficients-based features are widely used because of their superior performance (Carrión-Ojeda et al., 2021; Virgilio et al., 2020; Zarei and Asl, 2021). Accordingly, in this work, we combined DWT and OVR-CSP to develop a

novel FE method for classifying olfactory EEG signals. The proposed method extracted discriminative features by localizing the EEG signal in a time-frequency-space domain. Initially, our method conducted a multi-level wavelet decomposition on olfactory EEG signals of each channel using DWT. Subsequently, the multi-channel wavelet coefficients obtained at a specific level were spatially filtered using OVR-CSP. Following the spatial filtering, its variance was calculated to generate a feature vector, labeled as Wavelet Spatial Domain Feature (WSDF). In WSDF, we performed a five-level wavelet decomposition on EEG signals and adopted a Daubechies-4 (db4) as the mother wavelet function. Moreover, the detailed coefficients at the second level were selected to input OVR-CSP for spatial filtering.

The primary objective of this study is to design a novel feature that takes into account the time-varying and spatial characteristics of multi-channel EEG signals, to identify different olfactory stimuli effectively. To this end, two olfactory EEG datasets, including a public dataset called "Odor Pleasantness Perception Dataset" (OPPD) (Kroupi et al., 2014) and our self-collected dataset, were utilized to validate the effectiveness of the proposed method. The public OPPD dataset was collected from five healthy male participants, who experienced four odor stimuli in the eyes-open and eyes-closed conditions. For our self-collected dataset, the EEG signals were recorded as the 11 healthy subjects (8 males and 3 females) with eyes open smelled thirteen different odors. Based on two EEG datasets, we compared the proposed WSDF with DWT coefficients features, the spatial-domain feature (i.e., OVR-CSP) and other EEG features characterizing odors. Moreover, we also compared our results with previous studies, which use the same dataset.

The major innovations and contributions of this work are as follows. 1) An advanced feature (i.e., WSDF) was developed for decoding olfactory EEG signals. To the best of our knowledge, this is the first study, where WSDF has been proposed and utilized to distinguish different olfactory stimuli from EEG signals. 2) In order to illustrate the benefits of the proposed WSDF, we conducted a range of comparative studies based on two datasets i.e., a public OPPD dataset and our self-collected dataset. In particular, a wide range of features, including time-domain, frequency-domain, time-frequency-domain, spatial-domain and nonlinear features, were systematically compared on identical olfactory EEG datasets. Lastly, experimental results revealed a superior classification performance for our proposed WSDF method.

The remainder of this paper is organized as follows. Section 2 introduces the methods and materials, including olfactory decoding model, the DWT, OVR-CSP and WSDF algorithms, EEG datasets, and evaluation procedure. Results and discussion are presented in Section 3. Finally, the work is concluded in Section 4.

2. Methods and materials

2.1. Olfactory decoding model

Fig. 1 shows the block diagram of decoding olfactory EEG signals based on the proposed WSDF in this study. It consists of four stages: EEG collection, preprocessing, WSDF extraction and classification.

In the preprocessing stage, the collected multi-channel olfactory EEG signals were filtered by a bandpass filter, which can enhance the SNR of EEG signals. In the WSDF extraction stage, DWT was used to perform a multilevel wavelet decomposition on the preprocessed EEG signals, and then the multi-channel wavelet coefficients obtained at a specific level were fed into OVR-CSP for spatial filtering and FE. Finally, support vector machine (SVM) was utilized as a classifier to classify olfactory EEG signals.

2.1.1. Discrete wavelet transform (DWT)

DWT can effectively capture and locate the local characteristics of nonstationary signals in both time and frequency domains. It has been widely used in the analysis of EEG signals (Carrión-Ojeda et al., 2021; Virgilio et al., 2020; Zarei and Asl, 2021).

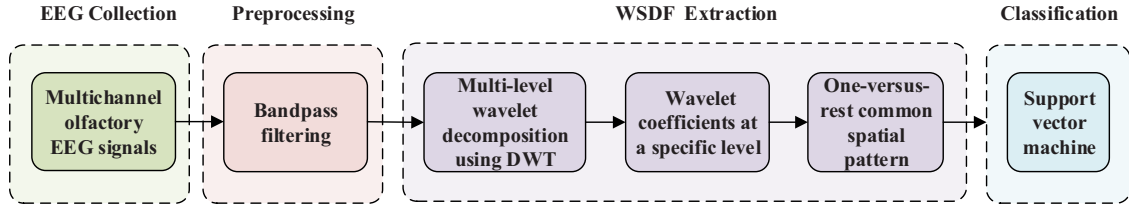


Fig. 1. The block diagram of decoding olfactory EEG signals based on the proposed feature extraction method.

Given a single-channel EEG signal $x(n)$ ($n = 1, 2, 3, \dots, T$), sampling frequency fs , DWT can perform a J -level ($J = 1, 2, 3, \dots, \log_2(T)$) decomposition on it by utilizing a scaling and a wavelet function associated with low-pass and high-pass filters. At each decomposition level, the obtained wavelet coefficients are recorded as approximation coefficients (Eq. (1)) and detail coefficients (Eq. (2)). Approximation coefficients represent low frequencies of the signal, corresponding to the frequency range $0 \sim fs/2^{j+1}$ (j represents the current decomposition level). Detail coefficients, on the other hand, represent high frequencies of the signal, corresponding to the frequency range $fs/2^{j+1} \sim fs/2^j$.

$$cA_j(n) = \sum g(k) cA_{j-1}(2n-k) \quad (1)$$

$$cD_j(n) = \sum h(k) cA_{j-1}(2n-k) \quad (2)$$

Initially, $cA_0(n) = x(n)$, $cA_j(n)$ and $cD_j(n)$ represent the approximation and detail coefficients of the j th level, respectively. $g(k)$ is a low pass filter and $h(k)$ is a high pass filter, which are respectively calculated by (3) and (4) as follows:

$$g(k) = \frac{1}{\sqrt{2}} \int \varphi\left(\frac{t}{2}\right) \varphi^*(t-k) dt, \quad (3)$$

$$h(k) = \frac{1}{\sqrt{2}} \int \psi\left(\frac{t}{2}\right) \varphi^*(t-k) dt. \quad (4)$$

Here, $\varphi(t)$ is the scaling function, $\psi(t)$ is the wavelet function and $\varphi^*(t-k)$ is the conjugation of $\varphi(t-k)$.

2.1.2. One-versus-rest common spatial pattern (OVR-CSP)

As a spatial-domain feature extractor, the fundamental principle of CSP is to construct a projection matrix that is composed of several pairs of spatial filter vectors, to maximize the variance of two-class signals. For multi-channel EEG signals, CSP can reduce its dimensionality automatically and extract its spatial features with lower computational burden.

OVR-CSP is an extension of CSP from two-class to a multi-class case. Assume that there are k -class ($k \geq 3$) multi-channel EEG signals $\mathbf{X}_{ij} \in \mathbb{R}^{r \times T}$, $\mathbf{X}_{ij}$ represents the j -th trial of the i -th class, r is the number of channels, and T is the number of sampling points. OVR-CSP divides this k -classification into a number of k one-to-one two-classification, and constructs a spatial filter for each two-classification. The specific steps are described as follows:

Step 1. : First, select the EEG signal belonging to a certain class as one group ($\mathbf{G}_{1,1}, \dots, \mathbf{G}_{1,N} \in \mathbb{R}^{r \times T}$) from the k -class EEG signals, and the remaining $k-1$ class EEG signals as the other group ($\mathbf{G}_{2,1}, \dots, \mathbf{G}_{2,M} \in \mathbb{R}^{r \times T}$). Here, N and M are the total trials of each group, $\mathbf{G}_{ij}$ represents the j -th trial of the i -th group. Then, the mean covariance matrices of two groups of EEG signals are calculated as follows:

$$\bar{\mathbf{C}}_1 = \frac{1}{N} \sum_{j=1}^N \frac{\mathbf{G}_{1,j} \mathbf{G}_{1,j}^T}{\text{trace}(\mathbf{G}_{1,j} \mathbf{G}_{1,j}^T)}, \quad \bar{\mathbf{C}}_2 = \frac{1}{M} \sum_{j=1}^M \frac{\mathbf{G}_{2,j} \mathbf{G}_{2,j}^T}{\text{trace}(\mathbf{G}_{2,j} \mathbf{G}_{2,j}^T)}, \quad (5)$$

where $\mathbf{G}^T$ is the transpose of $\mathbf{G}$, and $\text{trace}(\mathbf{G} \mathbf{G}^T)$ is the trace of $\mathbf{G} \mathbf{G}^T$.

Step 2. : Calculate the mixed covariance matrix $\mathbf{C}$, and perform

eigenvalue decomposition on it:

$$\mathbf{C} = \bar{\mathbf{C}}_1 + \bar{\mathbf{C}}_2 = \mathbf{U} \boldsymbol{\lambda} \mathbf{U}^T, \quad (6)$$

where $\boldsymbol{\lambda}$ is the diagonal matrix of eigenvalues, and $\mathbf{U}$ is the corresponding eigenvector matrix.

Step 3. : Obtain a whitening matrix $\mathbf{W}$ by utilizing whitening transformation:

$$\mathbf{W} = \sqrt{\boldsymbol{\lambda}^{-1}} \mathbf{U}^T, \quad (7)$$

and then two matrices $\mathbf{S}_1$ and $\mathbf{S}_2$ can be given as:

$$\begin{cases} \mathbf{S}_1 = \mathbf{W} \bar{\mathbf{C}}_1 \mathbf{W}^T = \mathbf{U}_s \boldsymbol{\lambda}_1 \mathbf{U}_s^T \\ \mathbf{S}_2 = \mathbf{W} \bar{\mathbf{C}}_2 \mathbf{W}^T = \mathbf{U}_s \boldsymbol{\lambda}_2 \mathbf{U}_s^T \end{cases}, \quad (8)$$

where $\mathbf{S}_1$ and $\mathbf{S}_2$ share the same eigenvectors. In particular, the sum of $\boldsymbol{\lambda}_1$ and $\boldsymbol{\lambda}_2$ is the identity matrix. Therefore, when the eigenvalue of $\mathbf{S}_1$ is the largest, the corresponding eigenvalue of $\mathbf{S}_2$ is the smallest, and the covariance difference between two groups of signals is the largest.

Step 4. : The eigenvalues are then sorted in descending order. Thus, for the certain class of EEG signals selected in step 1), its CSP projection matrix $\mathbf{Q}$ can be defined as:

$$\mathbf{Q} = \mathbf{U}_s^T \mathbf{W} \in \mathbb{R}^{r \times r}. \quad (9)$$

The first m ($1 \leq m \leq r/2, m \in \mathbb{Z}$) and the last m rows of the matrix $\mathbf{Q}$ can be extracted to construct a new filter $\mathbf{P} \in \mathbb{R}^{2m \times r}$, which aims to enhance features and reduce dimensionality of the original data.

Step 5. : Repeat steps 1–4 until the spatial filter $\mathbf{P}_c \in \mathbb{R}^{2m \times r}$ ($c = 1, 2, 3, \dots, k$) for each class of EEG signals is constructed.

Step 6: Input EEG signals $\mathbf{X}_{ij} \in \mathbb{R}^{r \times T}$ into the spatial filter $\mathbf{P}_c$ ($c = 1, 2, 3, \dots, k$) to obtain the corresponding projection matrix $\mathbf{Z}_c$:

$$\mathbf{Z}_c = \mathbf{P}_c \mathbf{X}_{ij} \in \mathbb{R}^{2m \times T}, c = 1, 2, 3, \dots, k. \quad (10)$$

Step 7: The logarithm of the normalized variance of each row in the matrix $\mathbf{Z}_c$ ($c = 1, 2, 3, \dots, k$) is calculated to obtain a feature vector denoted by:

$$\mathbf{V}_c = \log \left(\text{var} \left(\mathbf{Z}_c(i) \right) / \sum_{i=1}^{2m} \text{var}(\mathbf{Z}_c(i)) \right), c = 1, 2, 3, \dots, k, \quad (11)$$

where the symbol var represents the variance and $\mathbf{Z}_c(i)$ indicates the i -th row vector of $\mathbf{Z}_c$. The obtained $\mathbf{V}_c \in \mathbb{R}^{2m \times 1}$ ($c = 1, 2, 3, \dots, k$) is concatenated into a feature vector $\mathbf{V} = [\mathbf{V}_1^T, \mathbf{V}_2^T, \dots, \mathbf{V}_k^T]^T \in \mathbb{R}^{1 \times 2mk}$, where $\mathbf{V}_c^T$ is the transpose of $\mathbf{V}_c$ ($c = 1, 2, 3, \dots, k$). Finally, the feature vector $\mathbf{V}$ is used as the feature of EEG signals $\mathbf{X}_{ij}$. It should be noted that the dimension of the feature vector $\mathbf{V}$ is $2mk$. Here, k is the number of categories to be classified. It is a fixed value for a given classification task. While m is a variable parameter that has to be set by the experimenter.

2.1.3. Wavelet-spatial domain feature (WSDF)

Considering that DWT can incorporate the information from both time and frequency domains of EEG signals, and OVR-CSP can reflect the

spatial information of EEG signals, therefore, we combine DWT with OVR-CSP to localize the EEG signal in the time-frequency-space domain and extract the discriminative WSDF for classifying olfactory EEG signals. To extract the WSDF from multi-channel olfactory EEG signals, there are two core steps:

- 1) First, for the preprocessed multi-channel EEG signals $\mathbf{X}$, represented as a matrix of size $r \times T$, where r is the number of channels and T is the number of sampling points, a J -level decomposition is performed on each channel signal of $\mathbf{X}$ using DWT. A total of $J + 1$ wavelet coefficient matrices with different sizes are obtained, including an approximation coefficients matrix and J detail coefficients matrices.
- 2) A certain wavelet coefficients matrix selected from $J + 1$ wavelet coefficients matrices is used to replace multi-channel EEG signals $\mathbf{X}$. Then the selected wavelet coefficients matrix is put into OVR-CSP to implement spatial filtering and extract features. The resulting feature vector $\mathbf{V} \in \mathbb{R}^{1 \times 2mk}$ is the WSDF of $\mathbf{X}$.

2.2. Datasets description

2.2.1. Public dataset

A publicly-available OPPD dataset (Kroupi et al., 2014) was used in our study. This dataset was collected from five healthy right-handed subjects (i.e., five men aged between 26 and 32) recruited from the Swiss Federal Institute of Technology (EPFL). Experiments were conducted in both eyes-closed and eyes-open conditions. In each condition, subjects experienced four odor stimuli (Valerian, Lotus, Cheese, and Rosewater). EEG signals were recorded using a standard 256-channel system. The sampling frequency was 250 Hz. An EGI's Geodesic EEG System (GES) 300 was used to record, amplify, and digitalize the EEG signals (Kroupi et al., 2014).

The public EEG data has been preprocessed by the provider. The filtering and preprocessing steps carried out by the dataset provider are explained as follows (Kroupi et al., 2014). First, the heavy artifacts of EEG signals were manually removed, resulting in a total of 216 electrodes for further processing. Then, in order to remove slow drifts (Yazdani et al., 2012), EEG data were filtered using a third-order Butterworth high-pass filter with a cut-off frequency of 3 Hz. Next, a FastICA algorithm was implemented to remove the eye-blinking artifacts of the EEG data recorded in the eyes-open condition. Finally, EEG data were re-referenced to the common average. More details can be found in Ref. Kroupi et al. (2014). In this study, in order to reduce the interference of high-frequency noise signals, the EEG data were further filtered between 3 and 70 Hz using a third-order Butterworth infinite impulse response filter.

2.2.2. Our dataset

The public OPPD dataset is a small-scale dataset that only has five subjects' EEG signals induced by four different olfactory stimuli. To further assess the robustness of our method, a larger dataset built by our laboratory was used.

In our experiment, a total of eleven healthy right-handed subjects (eight males and three females aged 24.9 ± 3.0 years old) participated in the EEG data collection. None of them had the psychiatric disorder, respiratory failure and neurological illness according to participants' self-reports. Before the experiment, participants were instructed about the purpose of the study and the experimental procedure. Every participant signed an informed consent form. The experimental process had been approved by the ethics committee of Tianjin University in China (ethical approval number: TJUE-2020-186). During the experiment, each subject was asked to open his/her eyes, relax, breath normally and avoid overt movements as much as possible. Thirteen odors were used as stimuli. Among of them are five olfactometer odors: rose, caramel, rotten smell, canned peaches, and excrement. The remaining eight odors were essential oils: mint, tea, coffee, rosemary, jasmine, lemon, vanilla, and lavender.

The experiment was conducted in a controlled room equipped with an air conditioner, a humidifier and an exhaust fan. The room temperature was around 26 °C. A Cerebus System was used to record, amplify, and digitalize EEG signals. It is designed by the Blackrock Microsystems, U.S., and consists of a front-end amplifier, a power supply, a Cerebus data acquisition equipment, and a workstation. According to the international 10–20 system, 32-channel EEG signals were recorded at a sampling rate of 1 kHz. The EEG signals from the A1 and A2 electrodes linked to mastoids were chosen as the references.

Our experiment consisted of thirteen runs for each subject, and each run corresponded to one odor, i.e., there were totally thirteen odors. For each run, the experimenter randomly chose an odor attached to an odorless filter strip and placed it under the subject's nose (1–2 cm under both nostrils) for ten seconds, which constituted a trial. This process was repeated 35 times, resulting in 35 trials per odor. Between two trials of the same odor, the time interval was set to four seconds in order to prevent adaptation and subject's fatigue (Kroupi et al., 2014).

After one run, the subject was given a 2-minute break. During this time, the door and the exhaust fan of the room were opened to evacuate the odor. After this break, another odor was randomly selected and the same procedure was repeated until all the thirteen odors were presented. Fig. 2 shows the detailed protocol. For each subject, there were 13 classes of EEG signals corresponding to 13 kinds of odors stimuli, and each class of EEG signals contained 35 trials. Thus, a total number of $13 \times 35 = 455$ EEG trials were obtained for each subject.

After the recording stage, EEG data were preprocessed by the EEGLAB toolbox on Matlab. First, two-channel EEG signals recorded at the reference electrodes A1 and A2 were excluded. Then, the remaining 30-channel EEG signals were filtered to remove the high frequency noise using a bandpass fifth-order Butterworth filter between 0.5 and 70 Hz (Hou et al., 2020b). Last, the filtered data were down-sampled to 256 Hz.

2.3. Evaluation procedure

Fig. 3 presents the overall flow chart for evaluating the performance of the proposed WSDF in classifying olfactory EEG signals.

As shown in Fig. 3, the preprocessed EEG data for each subject are split into a training set and a test set using the hold-out CV method or 5-fold CV method. In the training stage, the feature is obtained by extracting WSDF from the training set, and then it is used to train the SVM classifier. In the test stage, the trained SVM classifier predicts class labels based on the WSDF extracted from the corresponding test set. Finally, the classification accuracy of the test set is calculated by comparing the actual and predicted labels. To illustrate the superior performance of the proposed WSDF, we also compared it with other EEG features. When evaluating the classification performance of other features, it only needs to replace the WSDF extraction block in Fig. 3 with the algorithm of the corresponding feature.

2.3.1. WSDF extraction

In this work, for each dataset, the decomposition level J in the WSDF algorithm was set to 5 and the db4 wavelet was adopted as a mother wavelet function. As mentioned earlier, the sampling frequency of EEG signals is 250 Hz and 256 Hz in OPPD dataset and our self-collected dataset, respectively. A five-level wavelet decomposition is adopted for both datasets because it captures the dominant physiological EEG rhythms, i.e., delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), and gamma bands (30–70 Hz). In addition, the db4 wavelet used in this work is popular in the analysis of EEG signals (Alazrai et al., 2019; Carrión-Ojeda et al., 2021; Rajashekhar et al., 2021; Zarei and Asl, 2021). It can detect changes of EEG signals thanks to its smoothing feature (Subasi, 2007). Taking our self-collected dataset as an example, a five-level wavelet decomposition of EEG signals is presented in Fig. 4.

Depending on the Nyquist rule, the detectable frequency range is 0–128 Hz for our self-collected EEG signals with a 256 Hz sampling

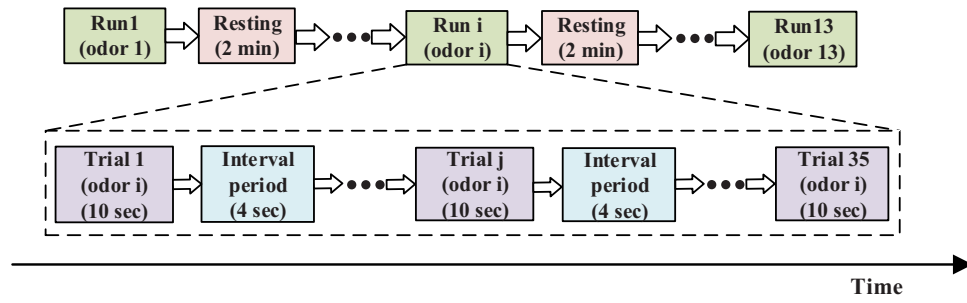


Fig. 2. The procedure of the olfactory-induced EEG experiment for each subject.

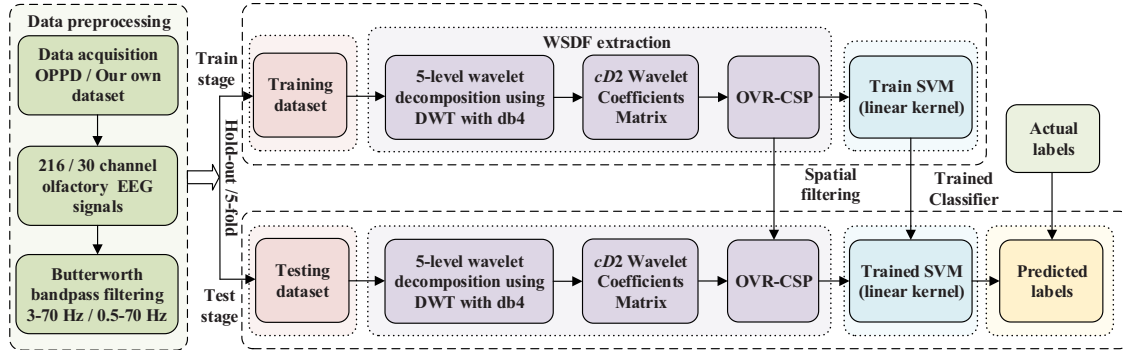


Fig. 3. The overall flow chart for evaluating the performance of the proposed WSDF in classifying olfactory EEG signals.

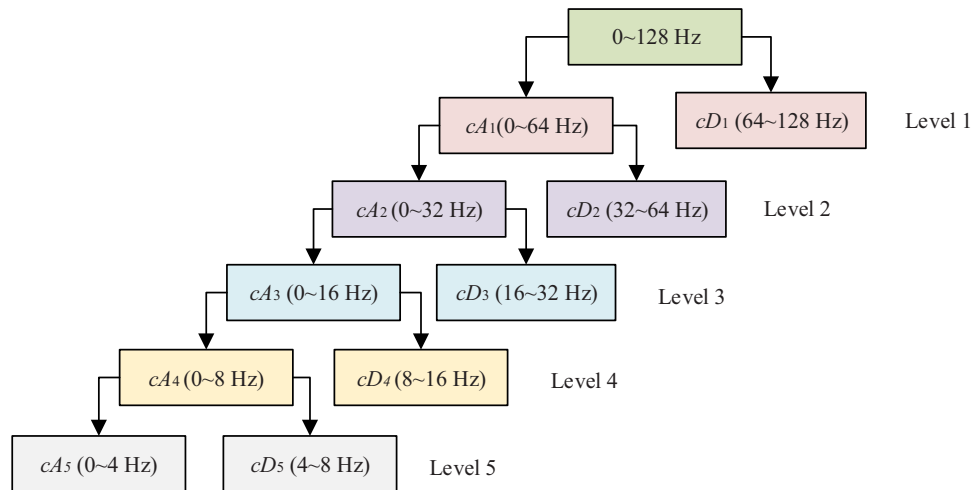


Fig. 4. Structure of a five-level wavelet decomposition of EEG signals with 256 Hz sampling frequency using DWT on our dataset.

frequency. Thus, the wavelet coefficients cA_5 , cD_5 , cD_4 , cD_3 , and cD_2 were extracted, corresponding to five main physiological EEG rhythms, i.e., delta, theta, alpha, beta, and gamma bands. Previous work has indicated that compared with other frequency bands (i.e., delta, theta, alpha, and beta bands), the gamma oscillation of brain activity is more associated with olfaction (Aydemir, 2017). Therefore, the wavelet coefficients matrix cD_2 , whose frequency band corresponds to the gamma band of physiological EEG rhythms, was employed in the extraction of WSDF.

2.3.2. Compared features

To demonstrate the effectiveness of the proposed WSDF in decoding olfactory EEG signals, we compared it with the spatial-domain feature OVR-CSP and DWT coefficients features, i.e., mean, variance and

entropy of DWT coefficients. In addition, WSDF was also compared with other EEG features that characterize odors (Aydemir, 2017, 2020; Ezzatdoost et al., 2020; Saha et al., 2014; Zhang et al., 2019). As shown in Table 1, these EEG features include three time-domain features (Aydemir et al., 2018; Aydemir, 2020; Saha et al., 2014), three frequency-domain features, a time-frequency-domain feature (Aydemir, 2017) and four nonlinear features (Ezzatdoost et al., 2020). Therefore, in this work, a total of 15 EEG features were used for comparison with our proposed WSDF.

All the EEG features listed in Table 1 were calculated according to the description in previous literatures (Aydemir, 2020, 2017; Ezzatdoost et al., 2020; Hou et al., 2020b; Saha et al., 2014; Zhang et al., 2019), except for the OVR-CSP and DWT coefficients-based features. To be specific, for time-domain features, Statistical measures (Aydemir, 2020)

Table 1

Five main categories of EEG features used for comparison with our proposed WSDF.

Feature Types	Extracted features
Time-domain features	1. Statistical measures: kurtosis, skewness, standard deviation, variance, and mean. 2. Hjorth parameters (Hjorth): activity, mobility, complexity.
Frequency-domain features	3. Autoregressive parameters (AR). 4. Power spectral density (PSD): mean. 5. Band power (BP). 6. Differential entropy (DE).
Time-frequency-domain features	7. Mean of discrete wavelet transform coefficients (MDC). 8. Variance of discrete wavelet transform coefficients (VDC). 9. Entropy of discrete wavelet transform coefficients (EDC). 10. The absolute values of continuous wavelet transform coefficients (CWTs): mean and standard deviation.
Nonlinear features	11. Higuchi's fractal dimension (HFD). 12. Largest Lyapunov exponent (LLE). 13. Approximate entropy (ApEn). 14. Lempel-Ziv complexity (LZC).
Spatial-domain features	15. One-versus-rest common spatial pattern (OVR-CSP).

includes the common kurtosis, skewness, standard deviation, variance, and mean. Three Hjorth parameters, namely Activity, Mobility and Complexity, are used to examine if the EEG signal is oscillatory or not (Oh et al., 2014). These parameters can be formulated as (Aydemir, 2020; Oh et al., 2014):

$$\text{Activity}(x(t)) = \frac{1}{T} \sum_{n=1}^L (x(t) - \mu)^2, \quad (12)$$

$$\text{Mobility}(x(t)) = \sqrt{\frac{\text{Activity}(\dot{x}(t))}{\text{Activity}(x(t))}}, \quad (13)$$

$$\text{Complexity}(x(t)) = \frac{\text{Mobility}(\dot{x}(t))}{\text{Mobility}(x(t))}, \quad (14)$$

where μ denotes the mean of the signal $x(t)$ and $\dot{x}(t)$ is the time derivative of $x(t)$. Given a signal $x(t)$, it can be written as an AR model, i.e., $x(t) = \sum_{i=1}^p a_i x(t-i) + e(t)$, where $a_i (i = 1, 2, \dots, p)$ stands for the AR parameter, p is the AR order, and $e(t)$ indicates a white noise. In this study, the order p was set to 5 (Aydemir, 2020).

For frequency-domain features, PSD, as the most popular EEG feature, is defined by:

$$P_x(f) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t) e^{-j2\pi ft}|^2 dt, \quad (15)$$

where T is the sample time of the signal $x(t)$ and f stands for the frequency. The BP features are calculated by fast Fourier transform (FFT) (Aydemir, 2020): $BP_L = \sum_L |X(k)|^2$, where $X(k)$ is the FFT coefficients of the signal $x(t)$, L represents the frequency band range. Differential entropy (Zheng and Lu, 2015) is calculated by $h(X) = \log \sqrt{2\pi e \sigma^2}$, where X follows the Gaussian distribution $N(\mu, \sigma^2)$, π and e are constants, and σ represents the standard deviation of a time series.

For time-frequency-domain features, CWTs include the average and the standard deviation of the absolute values of the wavelet transform coefficients for all the channels (Aydemir, 2017). For nonlinear features (Ezzatdoost et al., 2020; Namazi et al., 2020), HFD evaluates the self-invariant and self-similar structures of the signal. LLE is used to measure the chaotic behavior of EEG signals. ApEn, as an entropy algorithm, is used for regularity evaluation. LZC is a symbolic metric, which measures the rate of generation of new patterns along a sequence.

The detailed description and calculation process of these four nonlinear features (HFD, LLE, ApEn and LZC) can be found in Ref. Ezzatdoost et al. (2020).

OVR-CSP was directly extracted from the preprocessed multi-channel EEG signals. Three DWT coefficients features, i.e., the mean of DWT coefficients (MDC), the variance of DWT coefficients (VDC), and the entropy of DWT coefficients (EDC), were extracted from the obtained detail coefficients cD_2 , which are defined as follows:

$$\text{MDC} = \frac{1}{L} \sum_{n=1}^L S(n), \quad (16)$$

$$\text{VDC} = \sqrt{\frac{1}{L-1} \sum_{n=1}^L (S(n) - \text{MDC})^2}, \quad (17)$$

$$\text{EDC} = - \sum_{n=1}^L |S(n)|^2 \log(|S(n)|^2), \quad (18)$$

where $S(n)$ is DWT coefficients and L is its corresponding length.

2.3.3. Classifier and cross-validation

In this study, a subject-specific classification scheme was adopted for the classification of olfactory EEG signals, i.e., the training and test data were from the same subject, excluding the EEG data of other subjects. SVM was used as the classifier on account of its superior performance in the olfactory EEG-based classification (Hou et al., 2020a). This study employed LIBSVM toolbox (Chang and Lin, 2011) to implement the SVM classifier with a linear kernel. The penalty factor c was determined by the grid search at a space $2^{[-10:10]}$ with a step of one. The index of classification accuracy, which is defined as the ratio of correct predictions to the total number of test trials, was used to evaluate the performance of the classifier in this study.

To ensure that our experimental results are reliable and stable, two CV methods, i.e., the hold-out CV and 5-fold CV methods, were utilized to train and test the classifier on each dataset. For each CV method, we carried out 10 times independently and took the average of the 10 results as the final classification accuracy to improve the robustness of the results.

In the hold-out CV method, in order to make sure that our results can be fairly compared with existing studies on the same dataset, different data division strategies were utilized for OPPD dataset and our own dataset. To be specific, for OPPD dataset, approximately half of the trials of each odor were randomly selected as the training set and the rest was the test set (Aydemir, 2017). For our own dataset, twenty trials of each odor were randomly selected as the training set and the rest fifteen trials were set as the test set (Hou et al., 2020b). Thus, in OPPD dataset, the ratio of training and test data is approximately 1:1, while it is 4:3 in our own dataset. In the 5-fold CV method, for each dataset, all the EEG trials on the same subject were first randomly divided into five equal subsets, and then four subsets were used for training and the remaining one for testing. This procedure was repeated 5 times so that each subset served once for testing. Thus, the ratio of training and test data is 4:1 in each dataset.

3. Results and discussions

In this section, the effectiveness of the proposed method was evaluated and discussed based on the public OPPD dataset and our own dataset.

3.1. Classification results on the public OPPD dataset

In the OPPD-based classification experiment, the EEG signals of five subjects were investigated in both eyes-open and eyes-closed conditions. A four-way classification was carried out using hold-out CV and 5-fold

CV methods. Using the SVM classifier trained based on different EEG features and CV methods, the classification results for the eyes-open and eyes-closed conditions are presented as box-plots in Figs. 5 and 6, respectively. In the subgraph of Figs. 5 and 6, each box-plot represents the classification accuracy for the five subjects, i.e., five values per box-plot.

One may notice from Fig. 5 that for the eyes-open condition, the proposed WSDF gives a perfect classification accuracy for each of five subjects. This means that all the four odors (Valerian, Lotus, Cheese, and Rosewater) were correctly recognized using WSDF for each subject with eyes open. To investigate whether the classification performance of WSDF is significantly different from that of other features, a paired sample *t*-test was applied to the classification accuracy obtained using WSDF and the 15 features listed in Table 1. The results of *t*-test illustrate that WSDF yields the highest average classification accuracy (100%), which is significantly better than the other features ($p < 0.05$).

For the eyes-closed condition, we obtain from Fig. 6(a) that when using the hold-out CV method, WSDF gives an average classification accuracy of $90.73\% \pm 10.90\%$, which is inferior to the $92.94\% \pm 7.33\%$ provided by PSD. However, the difference of classification performance between WSDF and PSD is not statistically significant ($p > 0.05$). Moreover, the results in Fig. 6(b) indicate that when using the 5-fold CV method, a maximum average classification accuracy of $94.47\% \pm 8.16\%$ is obtained by WSDF (but no significant difference compared to the PSD with a $93.50\% \pm 9.68\%$ average classification accuracy, $p > 0.05$). Fig. 7 represents the average confusion matrix obtained by SVM applied to WSDF across five subjects with eyes closed. As observed from Fig. 7, the Cheese and Rosewater odors can be recognized with high accuracies, whereas the Valerian and Lotus odors are often misclassified.

3.2. Classification results on our own dataset

In order to further verify the effectiveness and robustness of the proposed WSDF, a subject-specific thirteen-way classification experiment was carried out on our self-built dataset to discriminate thirteen olfactory stimuli. As before, a SVM classifier was trained using both hold-out CV and 5-fold CV methods and applied for each feature.

The classification results are presented as box-plots in Fig. 8. To investigate whether the classification performance of the proposed

WSDF on the eleven subjects of our own dataset is significantly better than that of other features, a paired sample *t*-test was applied to the classification accuracy obtained using WSDF and the 15 EEG features listed in Table 1, as previously. The results reveal that WSDF significantly outperforms other 15 features in both CV methods ($p < 0.05$). The average classification accuracy of using WSDF is $99.45\% \pm 0.77\%$ and $99.50\% \pm 0.69\%$ in the hold-out CV and 5-fold CV methods, respectively. We also looked into the confusion matrix across subjects for WSDF. The results indicate that all the thirteen odors using in our own dataset can be recognized with high accuracies. Furthermore, it was noted from Fig. 8 that the classification performance of CWTs is second only to WSDF. It gives an average classification accuracy of $98.13\% \pm 2.11\%$ and $98.58\% \pm 1.73\%$ in the hold-out CV and 5-fold CV methods, respectively, which also verifies the validity of the CWT-based FE method for olfactory EEG classification (2017).

3.3. Effects of classifiers on the performance of WSDF

To investigate whether the superior classification performance of WSDF is independent of the classifier, four classical classifiers, i.e., *k*-nearest neighbor (*k*-NN), naive Bayesian (NB), extreme learning machine (ELM), and backpropagation neural networks (BPNN), were employed to classify olfactory EEG signals in this subsection. The classification procedure was the same as previously. For *k*-NN, the nearest neighbor ($k = 1$) and Euclidean distance were used. For ELM, the number of neurons in the hidden layer was set to 100, and the activation function was the sigmoid function. BPNN consisted of an input layer, two hidden layers and an output layer. In BPNN, the learning rate was set to 0.1, and the maximum number of training iterations was set to 5000. The classification results of *k*-NN, NB, ELM and BPNN trained using different features and CV methods on each dataset are presented in Fig. 9.

In Fig. 9, the first and second columns present the classification results of four classifiers trained using the hold-out CV and 5-fold CV methods, respectively. The classification results shown in the first, second and third rows of Fig. 9 are obtained on OPPD dataset with the eyes-open condition, OPPD dataset with the eyes-closed condition and our own dataset, respectively. One may notice from Fig. 9(a) and (b) that when using other four classifiers (i.e., *k*-NN, NB, ELM and BPNN), our

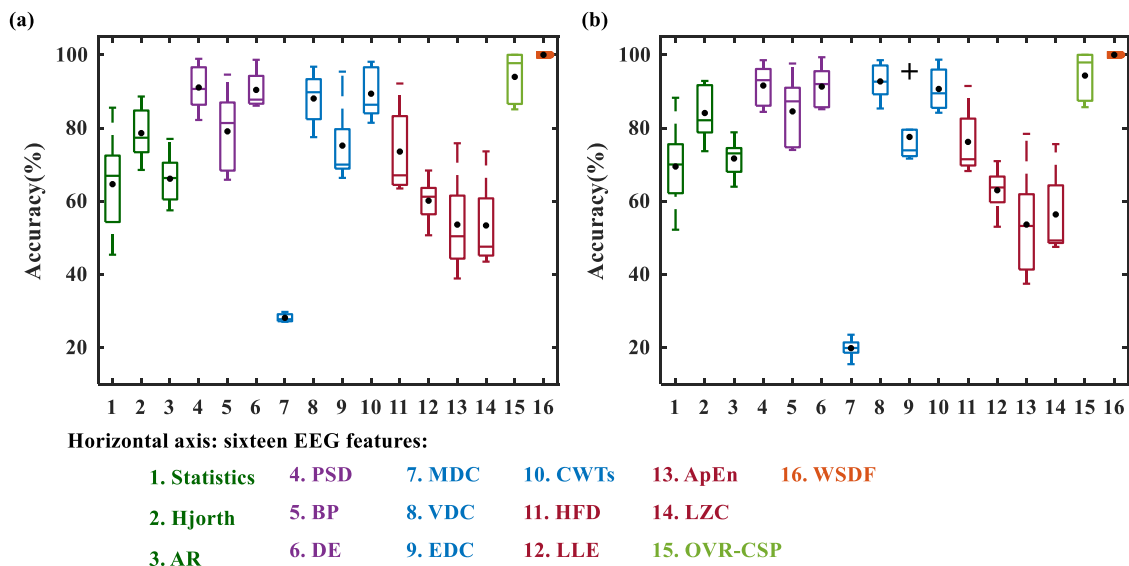


Fig. 5. Box-plots (the classification accuracy of five subjects) for each EEG feature on OPPD dataset with the eyes-open condition, when using the SVM classifier trained based on (a) the hold-out CV and (b) 5-fold CV methods. The numbers from 1 to 16 on the horizontal axis correspond to sixteen EEG features. For each box, the horizontal line is the median, the black dot is the mean, two edges respectively correspond to the 25th and 75th percentiles, the whisker is drawn from the ends of the interquartile ranges and extends to the most extreme data points not considered outliers, and the black crosses represent the most extreme data points that were considered outliers.

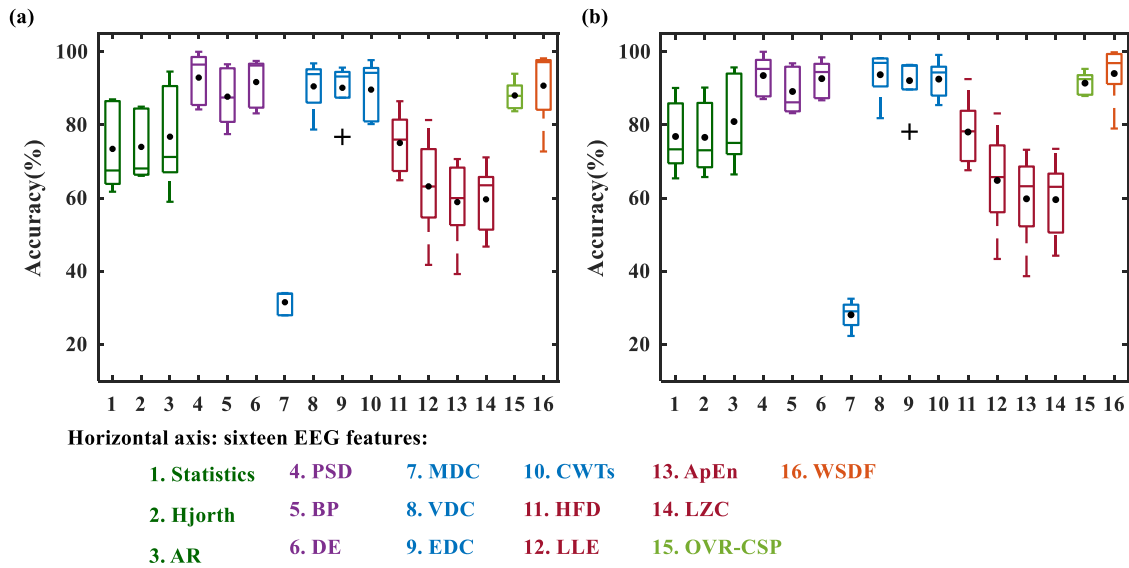


Fig. 6. Box-plots (the classification accuracy of five subjects) for each EEG feature on OPPD dataset with the eyes-closed condition, when using the SVM classifier trained based on (a) the hold-out CV and (b) 5-fold CV methods. The numbers from 1 to 16 on the horizontal axis correspond to sixteen EEG features. For each box, the horizontal line is the median, the black dot is the mean, two edges respectively correspond to the 25th and 75th percentiles, the whisker is drawn from the ends of the interquartile ranges and extends to the most extreme data points not considered outliers, and the black crosses represent the most extreme data points that were considered outliers.

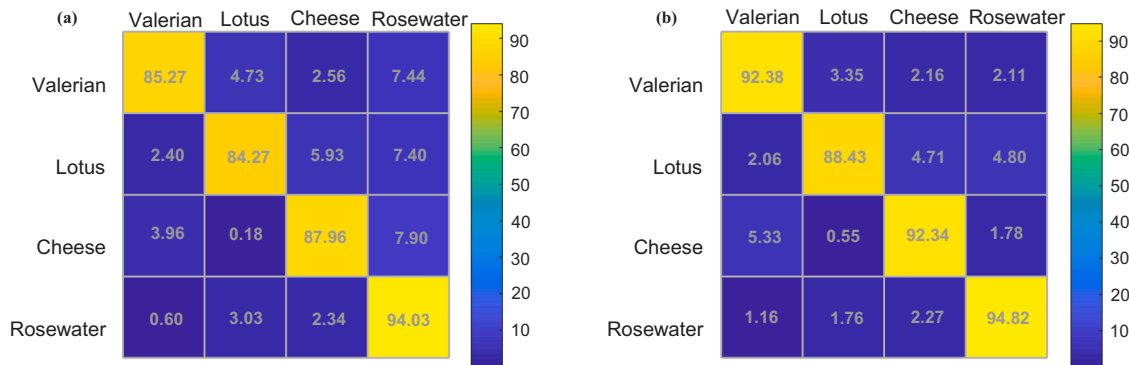


Fig. 7. For the eyes-closed condition in OPPD dataset, the average confusion matrix across five subjects by using our proposed WSDF and the SVM classifier trained based on (a) the hold-out CV method and (b) the 5-fold CV method.

proposed WSDF still gives a 100% average classification accuracy for the eyes-open condition in OPPD dataset, which also significantly outperforms other 15 EEG features ($p < 0.05$). For the eyes-open condition in OPPD dataset (Fig. 9(c) and (d)), similarly with the case in which SVM was used in Fig. 5, WSDF shows a high classification performance and is comparable with PSD for k -NN in both CV methods (no significant difference, $p > 0.05$). For Fig. 9(e) and (f), in all the four classifiers, the classification performance of using WSDF significantly outperforms those based on other features ($p < 0.05$). Therefore, we can conclude from the above results that the superior classification performance of the proposed WSDF is robust and is independent of the classifier used in this study.

3.4. Effects of parameter m on the performance of WSDF

According to the description in Section 2.1.3, we know that the parameter m in WSDF is a variable that needs to be manually set. It determines the dimensionality of the WSDF extracted from multi-channel olfactory EEG signals. For the four-way classification on OPPD dataset, the dimensionality of WSDF is $8m$, while it is $26m$ for the thirteen-way classification on our dataset. It can be seen that the larger the m , the higher the dimensionality of WSDF. In this subsection, we will

explore whether the parameter m affects the performance of WSDF in decoding olfactory EEG signals. If so, how does it affect and what is the optimal m ?

To solve the above problems, by using the WSDF with the parameter m varying from 1 to 10, we conducted the classification experiment of olfactory EEG signals as previously on each dataset. Figs. 10–12 present the classification results for the eyes -open condition in OPPD dataset, the eyes-closed condition in OPPD dataset, and our dataset, respectively. The most remarkable results in Figs. 10–12 are listed as follows:

- 1) In general, SVM and k -NN outperform the other three classifiers (i.e., NB, ELM, and BPNN). Moreover, on OPPD dataset (Figs. 10 and 11), the classification performance of SVM and k -NN are comparable (no significant difference between them, $p > 0.05$), while on our dataset (Fig. 12), SVM outperforms k -NN by 1–2% (statistically significant, $p < 0.05$).
- 2) The effect of parameter m on the performance of WSDF depends on the classifier. When using SVM or k -NN as the classifier, on both datasets, the performance of WSDF is not significantly ($p > 0.05$) improved as m increases from 2 to 10. However, when using NB, ELM or BPNN as the classifier, the effect of parameter m on the performance of WSDF is different on different datasets. For example, when

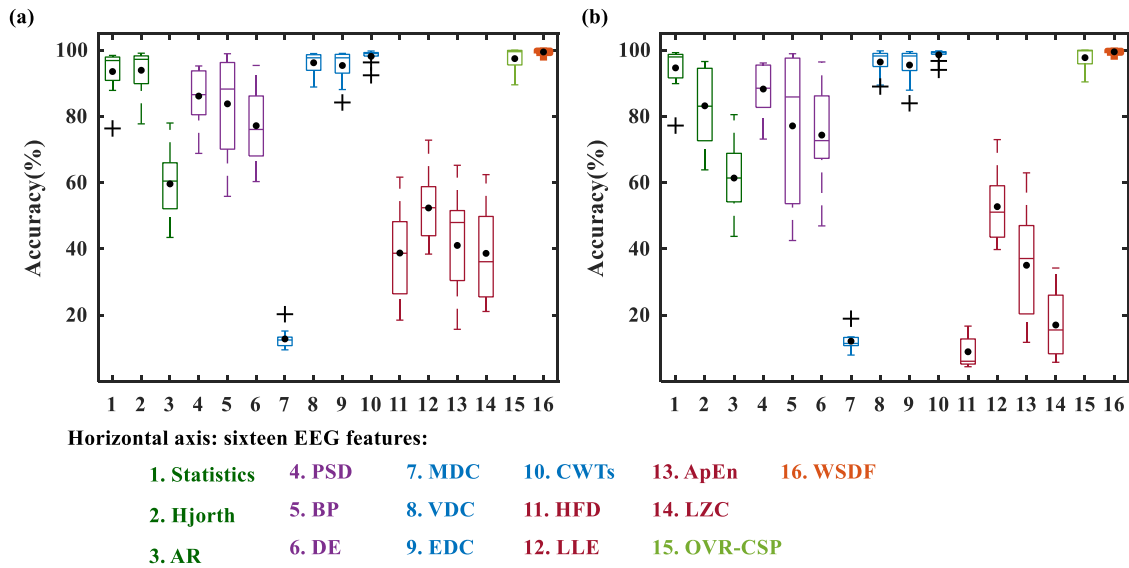


Fig. 8. Box-plots (the classification accuracy of eleven subjects) for each EEG feature on our own dataset, when using the SVM classifier trained based on (a) the hold-out CV and (b) 5-fold CV methods. The numbers from 1 to 16 on the horizontal axis correspond to sixteen EEG features. For each box, the horizontal line is the median, the black dot is the mean, two edges respectively correspond to the 25th and 75th percentiles, the whisker is drawn from the ends of the interquartile ranges and extends to the most extreme data points not considered outliers, and the black crosses represent the most extreme data points that were considered outliers.

using ELM as the classifier, the performance of WSDF increases first and then tends to be stable as m increases on OPPD dataset with eyes closed (Fig. 11), whereas on our dataset, its performance significantly decreases as m increases (Fig. 12).

- 3) Based on the experimental results obtained on two datasets, we can make the following conclusions: SVM is more suitable to classify olfactory EEG signals based on our proposed WSDF in term of classification accuracy. Moreover, setting the parameter m in WSDF to 2 is recommend when using SVM.

3.5. Compare to related work

In the research based on OPPD dataset (Aydemir, 2017, 2020; Ezzatdoost et al., 2020; Kroupi et al., 2014; Yazdani et al., 2012), to the best of our knowledge, only two works performed the same four-way classification experiment as we did in this study, as shown in Table 2. To be specific, Aydemir (2017) proposes a CWT-based FE method to distinguish different odor stimuli from EEG signals. By using the k -NN classifier trained based on the hold-out CV method, the average classification accuracies of $87.50\% \pm 4.3\%$ and $94.12\% \pm 2.9\%$ are achieved in the eyes-open and eyes-closed conditions, respectively. More recently, Ezzatdoost et al. (2020) introduce an olfactory stimulus decoder using nonlinear and chaotic features. They use the linear discriminant analysis (LDA) classifier trained based on the 5-fold CV method, and obtain the average accuracies of $96.71\% \pm 2.97\%$ and $88.79\% \pm 7.15\%$ in the eyes-open and eyes-closed conditions, respectively. Table 2 shows a comparison between the results of using our WSDF and the above-mentioned two works for classifying olfactory EEG signals on OPPD dataset.

From the results in Table 2, we can see that for the eyes-open condition, our method can accurately discriminate (i.e., the average accuracy of 100%) four different olfactory stimuli, which outperforms the previous work (Aydemir, 2017; Ezzatdoost et al., 2020). For the eyes-closed condition, the proposed method provides an average accuracy of $90.73\% \pm 10.90\%$ in the hold-out CV method, but it is lower than the average accuracy of $94.12\% \pm 2.9\%$ provided by Aydemir (2017). However, our method achieves an average accuracy of $94.06\% \pm 8.61\%$ in the 5-fold CV method, which is significantly better than the best accuracy of $88.79\% \pm 7.15\%$ reported by Ezzatdoost et al. (2020).

In addition, for the research based on our self-built olfactory EEG dataset, there is only one work to classify 13 different odors from EEG signals, as shown in Table 3. It can be seen from Table 3 that compared with the results reported by Hou et al. (2020b), the method proposed in this study significantly improves the classification accuracy.

From the abovementioned experimental results, we come to the conclusion that the proposed WSDF has great potential to distinguish different olfactory stimuli from EEG signals. One possible explanation for the superior performance of our proposed WSDF is that WSDF takes into account the temporal, frequency, and spatial characteristics of multi-channel EEG signals at the same time, rather than the single time, frequency, or time-frequency information.

3.6. Limits and future work

Despite the convincing ability of the proposed WSDF for decoding olfactory EEG signals, highlighted in this work, there are some limitations in the current study that can be addressed in future work.

From the results presented in Fig. 6 and Table 2, we noticed that for the eyes-closed condition in OPPD dataset, the proposed WSDF has no statistically significant advantage in terms of accuracy, compared with the frequency and time-frequency-domain features (e.g. PSD and CWTs). A possible reason for this is the lack of concentration of some subjects with eyes closed, which makes the spatial information of multichannel EEG signals, extracted through WSDF, indistinguishable for different olfactory stimuli. As described by the dataset provider (Kroupi et al., 2014), during the eyes-closed experiment, subjects were not asked to perform an explicit attention-demanding task, while they had to focus their attention during the eyes-open experiment. In addition, the provider reported that during odor perception, the brain activity of some subjects did not differ significantly for eyes-closed and eyes-open conditions (Kroupi et al., 2014). Therefore, there might exist a performance limitation for the proposed WSDF, when classifying the EEG signal collected from distracted subjects. With regards to this, a larger EEG dataset is required to further evaluate the performance of the proposed WSDF.

Moreover, some adjustments can be made in the experimental designs of both datasets (OPPD dataset and our dataset) to produce results that are more reliable. In the experiment, odor samples were manually presented to the subjects by an experimenter, and resultantly, some bias

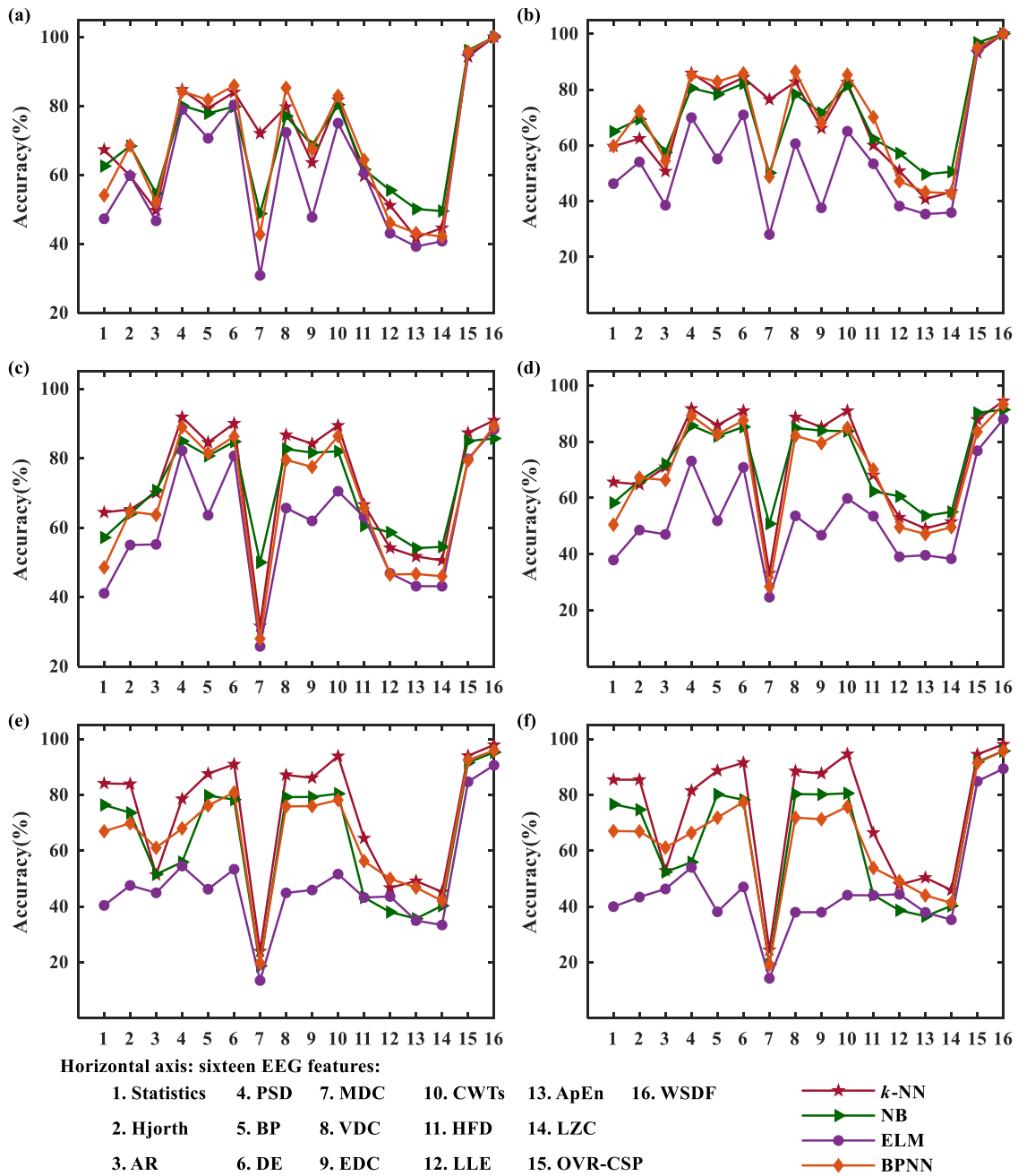


Fig. 9. Average classification accuracies of k -NN, NB, ELM and BPNN trained based on different features and CV methods. The first and second columns present the classification results of using the hold-out CV and 5-fold CV methods, respectively. The first, second and third rows present the classification results on OPPD dataset with the eyes-open condition, OPPD dataset with the eyes-closed condition and our own dataset, respectively. The numbers from 1 to 16 on the horizontal axis correspond to sixteen EEG features.

could be induced towards the stimuli. Thus, in the future work, an automated olfactory stimulation device, i.e., olfactometer (Lundström et al., 2010), will be used to carry out olfactory EEG experiments in a more controlled way. Based on olfactometer, the odor will be presented randomly for each trial instead of each run, to improve the effectiveness of olfactory stimulation and obtain higher-quality EEG signals.

To enhance the computational models for decoding olfactory EEG signals, a follow-up work will focus on investigating the EEG differences for different odors. Those include, but are not limited to, finding out the optimal frequency band modulation for different odors, identifying critical channels or brain areas associated with odor processing, and exploring whether the EEG activity of experiencing odors is different from that of odorless state.

4. Conclusion

In this study, we develop and validate a novel EEG feature (WSDF) based on the DWT and OVR-CSP algorithms for decoding olfactory EEG signals. Unlike the traditional feature design solely based on the time-domain, frequency-domain, or time-frequency-domain of EEG signals, the proposed WSDF integrates the time, frequency and spatial information of EEG signals. In WSDF, DWT was used to decompose the EEG signals and obtain wavelet coefficients. Later on, OVR-CSP was introduced to spatially filter wavelet coefficients and extract the discriminative WSDF. Based on two olfactory EEG datasets, i.e., the public OPPD dataset and our own dataset, the performance of the proposed WSDF was evaluated via different classifiers and CV methods. The results

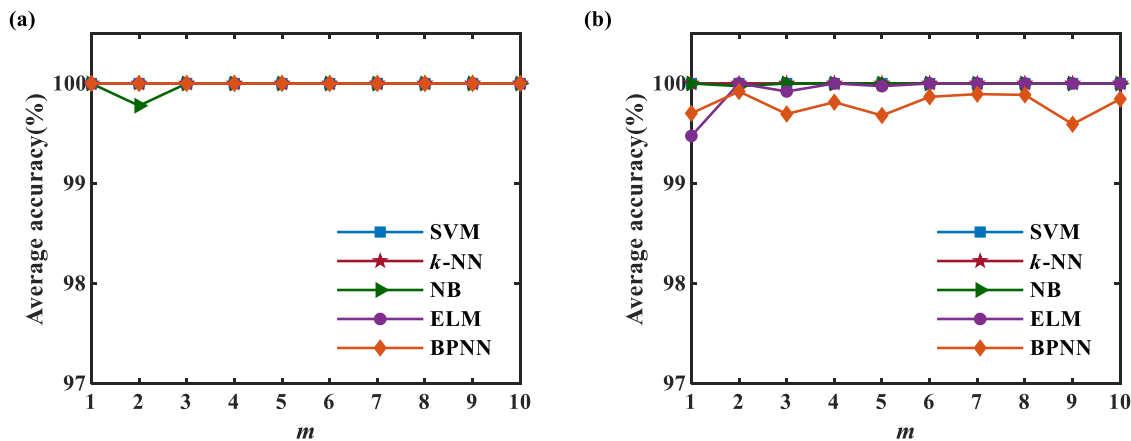


Fig. 10. For the eyes-open condition in OPPD dataset, the classification accuracy of using WSDF under different m for the classifier trained based on the (a) hold-out CV and (b) 5-fold CV methods.

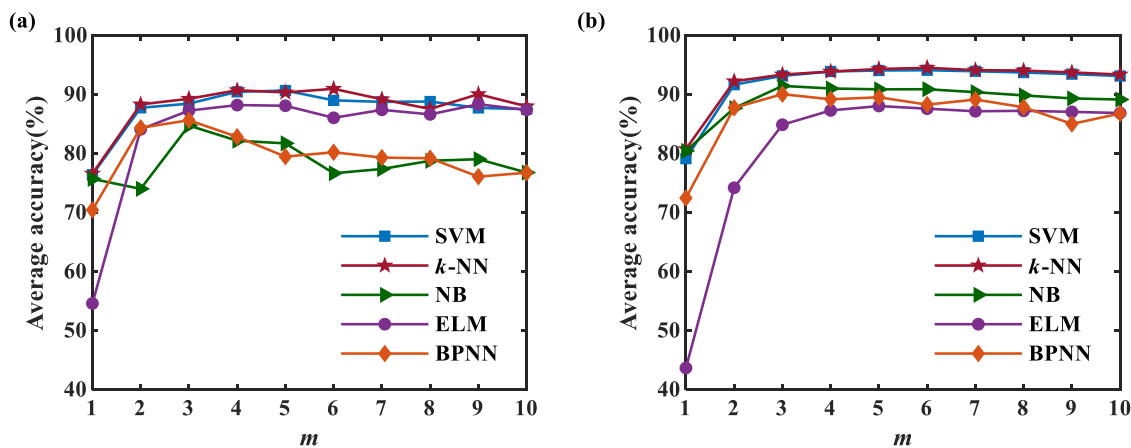


Fig. 11. For the eyes-closed condition in OPPD dataset, the classification accuracy of using WSDF under different m for the classifier trained based on the (a) hold-out CV and (b) 5-fold CV methods.

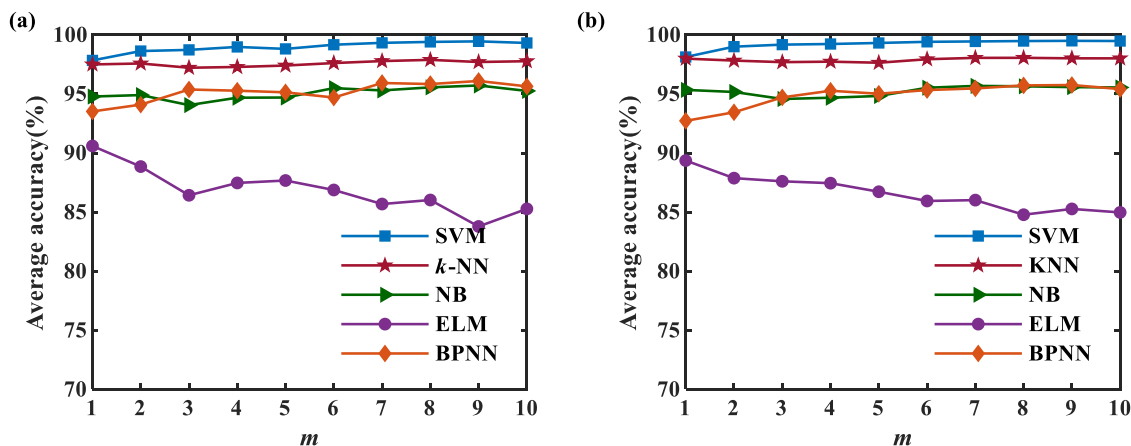


Fig. 12. For our own dataset, the classification accuracy of using WSDF under different m for the classifier trained based on the (a) hold-out CV and (b) 5-fold CV methods.

indicated that our proposed WSDF yielded the best classification accuracy of 100% and 94.47% on OPPD dataset for the conditions of eyes-open and eyes-closed, respectively. Likewise, the best classification accuracy for our own dataset turned out to be 99.50%. In conclusion, the proposed WSDF provides novel insights for the classification of olfactory EEG signals.

CRediT authorship contribution statement

Xiao-Nei Zhang: Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft, Writing – review & editing. **Qing-Hao Meng:** Supervision, Writing – review & editing, Resources, Funding acquisition. **Ming Zeng:** Conceptualization,

Table 2
Comparison of classification results between the proposed method and previous works on OPPD dataset.

References	Year	Eyes-Open (%)		Eyes-Closed (%)	
		Hold-out	5-fold	Hold-out	5-fold
Aydemir (2017)	2017	87.50 ± 4.3	–	94.12 ± 2.9	–
Ezzatdoost et al. (2020)	2020	–	96.71 ± 2.97	–	88.79 ± 7.15
This work	–	100 ± 0.0	100 ± 0.0	90.73 ± 10.90	94.47 ± 8.16

Table 3
Comparison of classification results between the proposed method and previous works on our own dataset.

References	Year	Hold-out (%)	5-fold (%)
Hou et al. (2020b)	2020	94.3 ± 4.2	92.9 ± 5.7
This work	–	99.45 ± 0.77	99.50 ± 0.69

Supervision, Writing – review & editing, Resources. **Hui-Rang Hou:** Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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